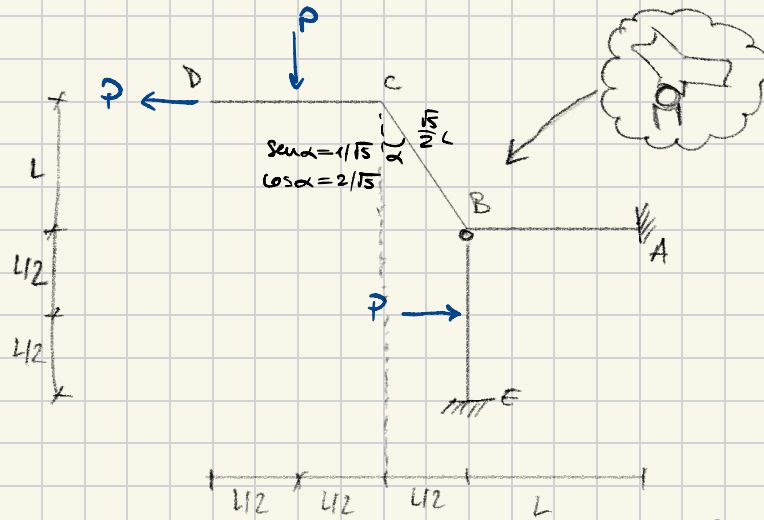
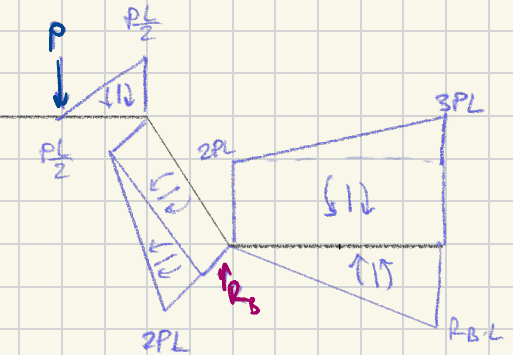
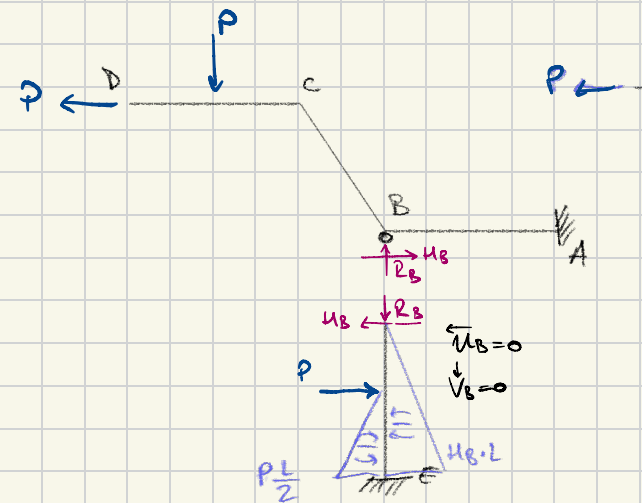


⊕ PROBLEMA 3.17

- 1) C.M. Trasl / Inters.
- 2) Diag. esfuerzos
- 3) Reacciones
- 4) Momentos en D



1) $CM = 2$, Traslac.



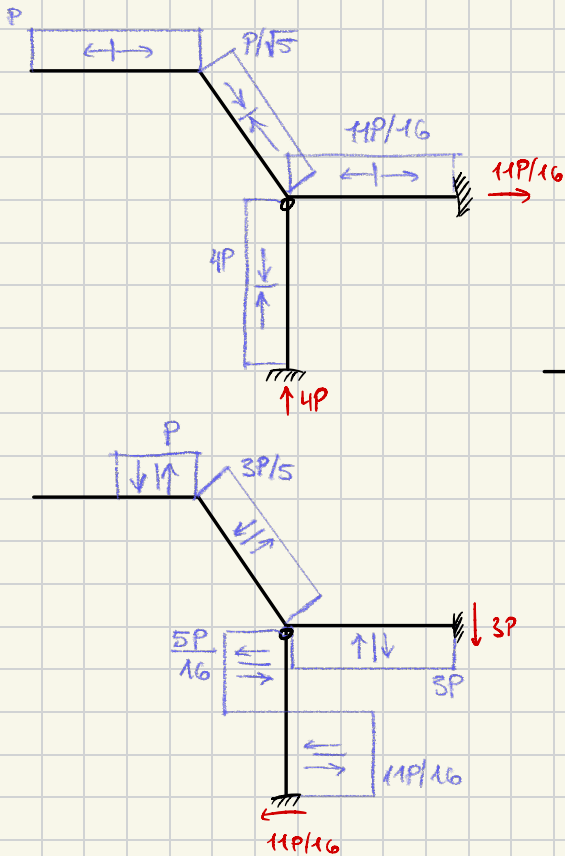
$$\downarrow V_B = \frac{1}{EI} \left[L \cdot 2PL \cdot \frac{1}{2} + \frac{1}{2} \cdot L \cdot PL \cdot \frac{2L}{3} - \frac{1}{2} L R_B L \cdot \frac{2L}{2} \right] = 0$$

$$R_B = 4P$$

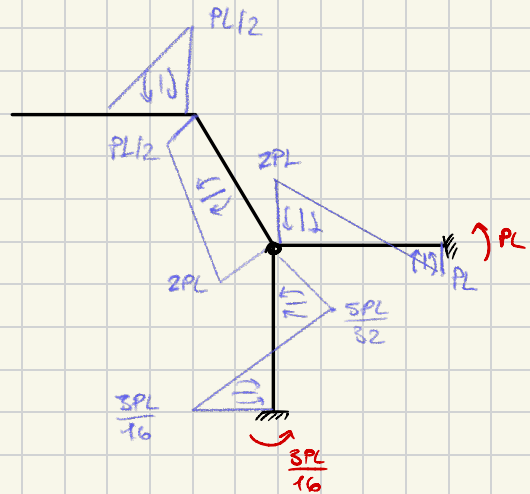
$$\uparrow M_B = 0 = \frac{1}{EI} \left[\frac{L}{2} \cdot H_B L \cdot \frac{2L}{3} - \frac{L}{4} \cdot \frac{PL}{2} \cdot \frac{5L}{6} \right] \Rightarrow$$

$$\Rightarrow H_B = \frac{5P}{16}$$

2,3) Diagr. de esf. y reacciones



$\text{Axiles} =$
 $P \sin \alpha - P \cos \alpha =$
 $(1/\sqrt{5} - 2/\sqrt{5})P =$
 $= -P/\sqrt{5}$
 $\text{Cortante} = P \cos \alpha + P \sin \alpha$
 $= 3P/\sqrt{5}$



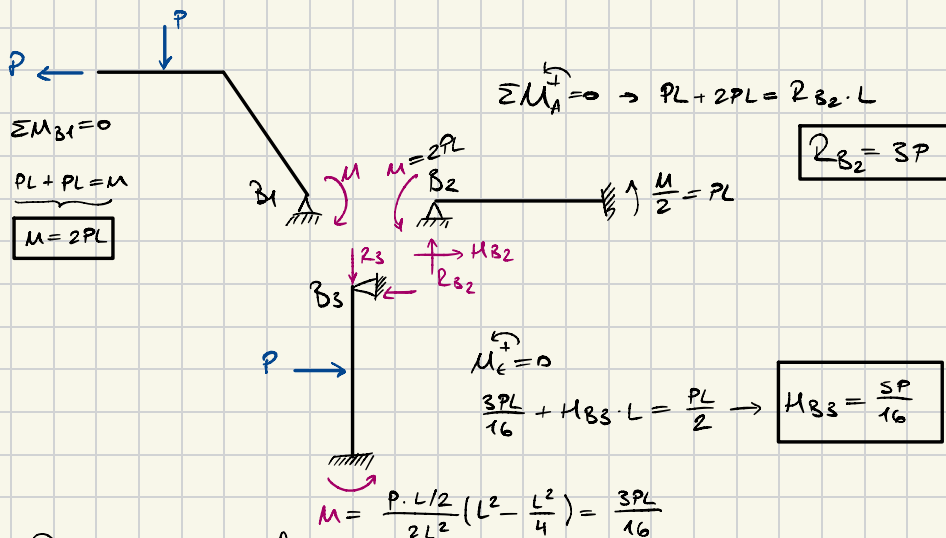
4. Movimientos en D

$$\bar{U}_D = \frac{1}{EI} \left[L \cdot 2PL \cdot L + \frac{L}{2} PL \cdot L - \frac{L}{2} 2PL \cdot L + \frac{\sqrt{5}}{2} \cdot L \cdot \frac{PL}{2} \cdot \frac{L}{2} + \frac{\sqrt{5}L}{4} \cdot \frac{3PL}{2} \cdot \frac{2L}{3} \right] = 1,34 \frac{PL^3}{EI}$$

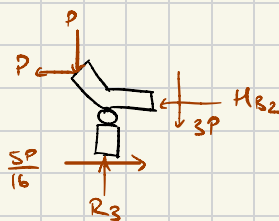
$$\bar{V}_D = \frac{1}{EI} \left[L \cdot 2PL \cdot 2L + \frac{L}{2} PL \left(L + \frac{L}{2} + \frac{2L}{3} \right) - \frac{L}{2} 2PL \cdot \left(L + \frac{L}{2} + \frac{2L}{3} \right) + \frac{\sqrt{5}}{2} L \cdot \frac{PL}{2} \cdot \left(L + \frac{L}{4} \right) + \frac{\sqrt{5} \cdot L}{4} \cdot \frac{3PL}{2} \left(L + \frac{2}{3} \frac{L}{2} \right) + \frac{L}{4} \cdot \frac{PL}{2} \left(\frac{L}{2} + \frac{2}{3} \frac{L}{2} \right) \right] = 2,67 \frac{PL^3}{EI}$$

$$\bar{\theta}_D = \frac{1}{EI} \left[L \cdot 2PL + \frac{L}{2} PL - \frac{L}{2} 2PL + \frac{\sqrt{5}}{2} L \cdot \frac{PL}{2} + \frac{\sqrt{5}L}{4} \cdot \frac{3PL}{2} + \frac{L}{4} \cdot \frac{PL}{2} \right] = 2,023 \frac{PL^2}{EI}$$

→ Habríamos tardado menos, si la hubiéramos tomado como INTRASCACIONAL.



Reacciones que faltan
 ↓
 Equilibrio del nodo:



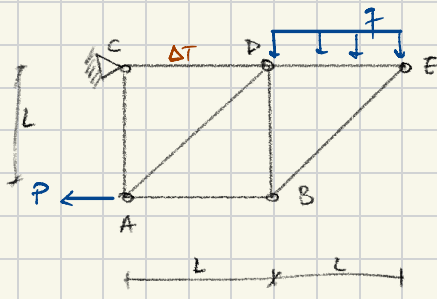
$\sum F_v = 0 \rightarrow P + 3P = R_3$
 $R_3 = 4P$

↑
 lo mismo que ANTES [OK ✓]

$\sum F_H = 0 \rightarrow P + H_{B2} - \frac{3P}{16} = 0$
 $H_{B2} = -\frac{11}{16} P$

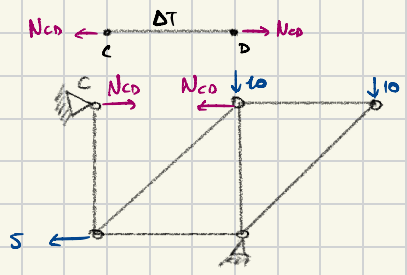
↑ ARTICULADA

* PROBLEMA INVENTADO



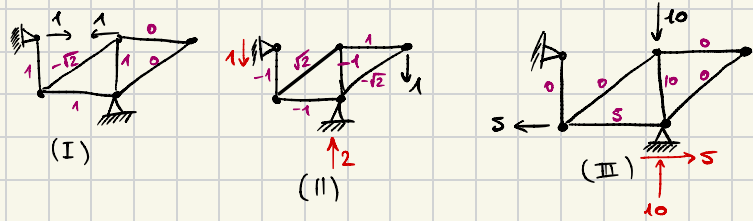
$$\begin{aligned} q &= 10 \text{ kN/m} \\ P &= 5 \text{ kN} \\ L &= 2 \text{ m} \\ EA &= 10^5 \text{ kN} \\ \Delta T &= 20^\circ\text{C} \\ \alpha &= 10^{-5} \text{ }^\circ\text{C}^{-1} \end{aligned}$$

$$1. - GH = 4 + 7 - 2 \cdot 5 = 1$$



$$\delta_{cd} = - \left[\frac{N_{cd} \cdot L}{EA} + \alpha \Delta T \cdot L \right]$$

$$N_i^0 = N_{cd} N_i^I + 10 N_i^{II} + N_i^{III}$$



$$\frac{\sum N_i^0 \cdot N_i^I \cdot L_i}{EA} = - \left[\frac{N_{cd} \cdot L}{EA} + \alpha \Delta T \cdot L \right]$$

$$\frac{1}{EA} \left[N_{cd} \sum N_i^{I^2} L_i + 10 \sum N_i^{II} \cdot N_i^I L_i + \sum N_i^{III} \cdot N_i^I L_i \right] = - \frac{N_{cd} L}{EA} + \alpha \Delta T \cdot L$$

$$\rightarrow N_{cd} = \frac{-10 \sum N_i^{II} \cdot N_i^I L_i - \sum N_i^{III} \cdot N_i^I L_i + \alpha \Delta T \cdot L \cdot EA}{\sum N_i^{I^2} L_i + L}$$

Se hace la tabla $\Rightarrow N_{cd} = (\dots)$

$$\downarrow V_D = \frac{\sum N_i^0 \cdot N_i^{II} L_i}{EA} = (\dots)$$